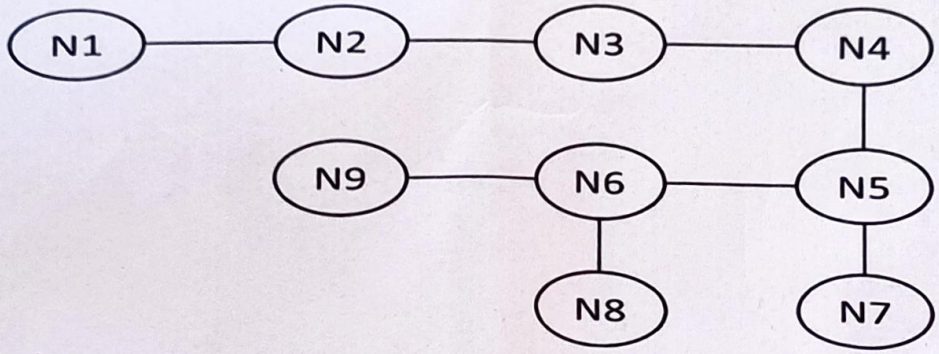
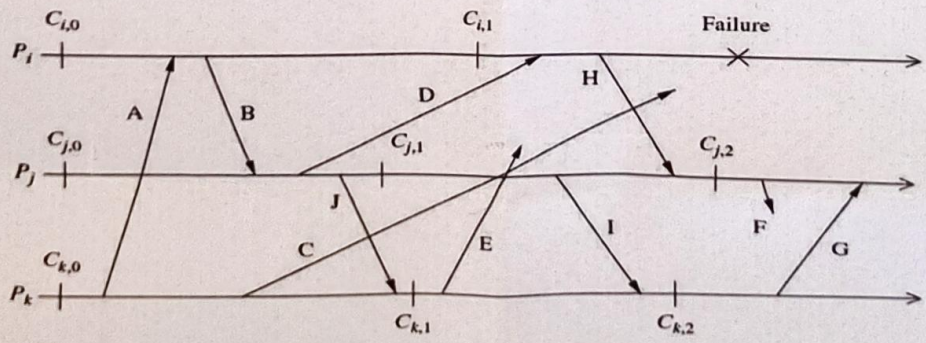
Notional Institute of Technology, Kurukshetra

(BTech 6th Sem), MST-II

Subject & code: DISTRIBUTED COMPUTING (CSPC- 304) **D.O.E: 08/04/2025**

Time: 50 mts **Marks: 15**

Attempt all the questions. All questions carry equal marks.

Q1.	Discuss Change-Robert's leader election algorithm on a ring of N nodes. Show that the algorithm requires $O(n^2)$ messages in the worst case.	(5)
Q2.	Let's consider the Raymond's spanning tree shown in Fig. 1. Node N3 holds the privilege.  **Fig.** 1. Based on the above said algorithm, answer the following questions. (i) Determine the values of the HOLDER variables of each of the nodes and redraw the spanning tree of Fig. 1. by replacing the undirected edges with directed edges. (ii) Illustrate, how mutual exclusion is guaranteed if Node N1 and N6 wants to enter into critical section, simultaneously, which does not hold the privilege.	(5)
Q3.	(a)What are the various limitations of synchronous checkpointing? Demonstrate how some of them have been addressed by asynchronous checkpointing? (b) For the diagram shown in **Fig.** 2. Find the global consistent state and list the orphan messages.  **Fig.** 2.	(5)